UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER
TIME	2026

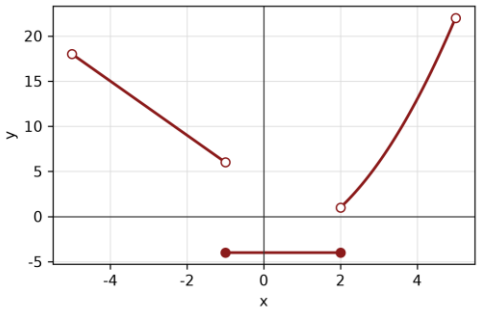
INSTRUCTIONS TO MARKERS

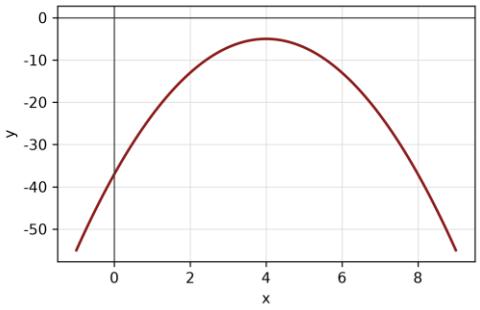
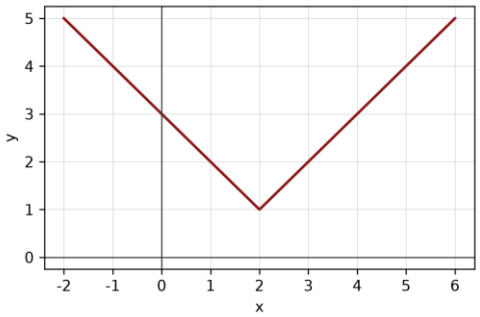
1. This marking scheme accompanies the MAT133 mock paper (November 2026).
2. Answers and marking points are for local study use only.

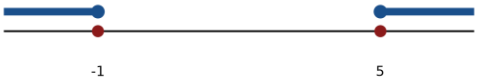
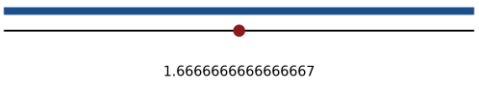
FOR MARKERS' USE ONLY

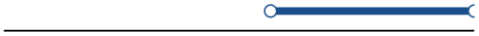
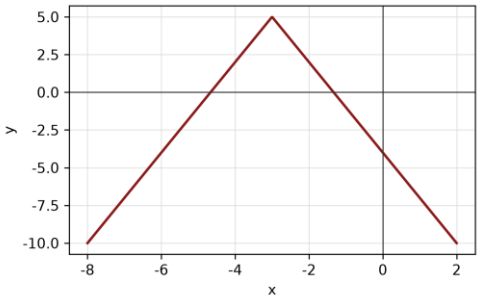
This marking scheme consists of 5 printed pages

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NO	SOLUTION		MARK
1 a)	$-1x + 2 \neq 0$	M1	4 m
	$x \neq 2$	A1	
	$2x + 6 \geq 0$	M1	
	$x \geq -3$	A1	
1 b)	$(g \circ f)(x) = g(f(x))$	M1	5 m
	$g(5 - 3x) = -\frac{5}{3x - 10}$	A1	
	$x \neq \frac{10}{3}$	B1	
1 c)	$y = \frac{3x + 2}{-4x - 5}$	M1	5 m
	$x(-4y + -5) = 3y + 2$	M1	
	$y(-4x - 3) = 2 - -5x$	A1	
	$f^{-1}(x) = \frac{-5x - 2}{4x + 3}$	B1	
1 d)	$f(x) = 3 - 3x, \quad x < -1$	B1	5 m
	$f(x) = -4, \quad -1 \leq x \leq 2$		
	$f(x) = x^2 - 3, \quad x > 2$		
	$f(-2) = 9, \quad f(0) = -4, \quad f(3) = 6$	A1	
			

NO	SOLUTION		MARK
1 e)	$y = -2(x-4)^2 - 5$ vertex $(4, -5)$ $y(0) = -37$ 	B1 B1 A1 A1	5 m
1 f)	$f(x) = 4x^5 - x^3 + 2x$ $f(-x) = -4x^5 + x^3 - 2x$ $-f(x) = -4x^5 + x^3 - 2x$ $f(-x) = -f(x)$	M1 A1 A1	4 m
1 g)	$(f+g)(x) = -7x - 4$ $(f-g)(x) = -3x - 8$ $(fg)(x) = 10x^2 + 2x - 12$	M1 A1 B1	4 m
1 h)	$y = \lvert x-2 \rvert + 1$ domain $[-2, 6]$ range $[1, 5]$ 	B1 B1 A1 A1	4 m

NO	SOLUTION	MARK
2 a)	$-x-8<1$ M1 $-x<9$ $(-9,\infty)$ A1  	4 m
2 b)	$-4<5-4x<3$ M1 $-9<-4x<-2$ M1 $\frac{1}{2}<x<\frac{9}{4}$ A1 	4 m
2 c)	$(-2x-2)(x-5)=0$ M1 $x=-1 \quad \text{or} \quad x=5$ A1 $(-\infty,-1]\cup[5,\infty)$ A1  	5 m
2 d)	$\frac{5-3x}{3x-5}+1>0$ M1 A1 False A1 $\varnothing$ A1  	5 m

NO	SOLUTION	MARK
2 e)	$3x - 6 < -2$ M1 $3x < 4$ $\left(-\infty, \frac{4}{3}\right)$ A1 	3 m
3 a)	$\lvert 5x + 1 \rvert = 5$ M1 $-5x - 1 = 5$ A1 $-5x - 1 = -5$ A1	4 m
3 b)	$\lvert 4x - 5 \rvert - 7 \geq 0$ M1 $5 - 4x >= 7$ or $5 - 4x <= -7$ A1 $\left(-\infty, -\frac{1}{2}\right] \cup [3, \infty)$ A1 	5 m
3 c)	$y = 5 - 3\lvert x + 3 \rvert$ B1 vertex $(-3, 5)$ A1 $\lvert x + 3 \rvert = \frac{5}{3}$ A1 	5 m